

PS Seminar 4

1. Algorithm in Dijkstra
2. Algorithm Bellman Ford
3. Shortest Path Faster Algorithm
4. Algorithm Roy Floyd
5. Matrix Multiplication

Algoritmul lui Dijkstra

$d[i]$ = distanța de la sursă la i

$d[\text{sursă}] = 0$, restul ∞

$d[x] + c(x, y) < d[y]$

$d[y] = d[x] + c(x, y)$

Model Examen 2022-2023 Subiect 04

1	2	3	4	5	6	7
∞	∞	0	∞	∞	∞	∞

1	2	3	4	5	6	7
3	10	0	∞	∞	∞	∞

1	2	3	4	5	6	7
3	10	0	∞	∞	4	∞

1	2	3	4	5	6	7
3	10	0	12	9	4	∞

1	2	3	4	5	6	7
3	10	0	11	9	4	∞

1	2	3	4	5	6	7
3	10	0	11	9	4	12

Complexitate

Burte foru : $O(N^2 + M) \approx O(N^2)$

Folosind Heap : $O(M \log N)$

Nu avem cicluri / muchii de cost negativ

Algorithmul Bellman Ford

for ($i = 1, N-1$)

Complexitate :

$O(N \cdot M)$

for (x, y, c) $\in E$

if $d[x] + c < d[y]$

$d[y] = d[x] + c$

Model Examen 2022-2023 Subiect 04

for ($i = 1, N-1$)

for (x, y, c) $\in E$

if $d[x] + c < d[y]$

$d[y] = d[x] + c$

for (x, y, c) $\in E$

după for

if $d[x] + c < d[y]$

Negative = True

Shortest Path Faster Algorithm

(SPFA)

Dacă am trecut prin relaxare

contorizam de câte ori adaugăm un nod în queue

$> n \Rightarrow$ ciclu negativ

$O(M)$

Algoritmă Roy Floyd

$d[i][j]$ = distanța minimă de la i la j

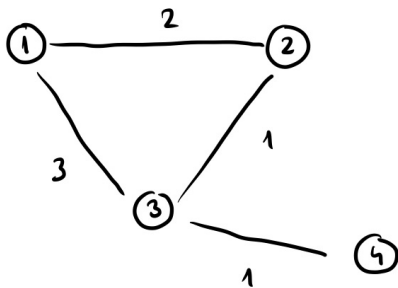
For $k = 1, n$

For $i = 1, n$

For $j = 1, n$

$$d[i][j] = \min (D[i][j], \\ D[i][k] + D[k][j])$$

D_0



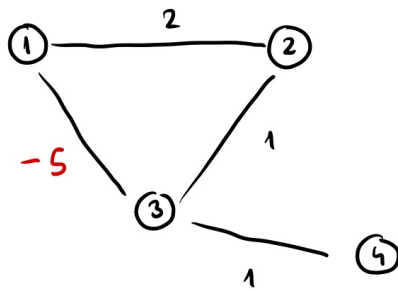
0	2	3	∞
2	0	1	∞
3	1	0	1
∞	∞	1	0

$D_k[i][j]$ = costul minim de la i la j

dacă putem folosi k noduri

intermedii, noduri din mulțimea $\{1, 2, \dots, k\}$

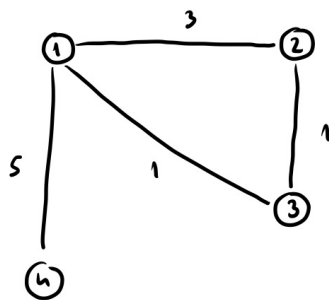
D_0



-2			
	0		
		0	
			0

Dacă avem pe diag. principală o val. negativă
 $\Rightarrow$ ciclu negativ

ex. Colloviu 2024-2025



	1	2	3	4
1	0	2	1	5
2	2	0	1	7
3	1	1	0	6
4	5	7	6	0

(i, j, k) = dacă există un drum de cost
 minim de la i la j care nu treacă prin k
 #

$(1, 2, 3) \rightarrow DA$

$(1, 2, 4) \rightarrow NU$

Complexitate unită $(N^3 + Q)$

Ref

1. Roy Floyd

2. Verif caso $\Delta[i][j] == \Delta[i][k] + \Delta[k][j]$

exemplar

$$(1, 2, 3) : \Delta[1][2] = 2$$

$$\Delta[1][3] + \Delta[3][2] = 2$$

" "

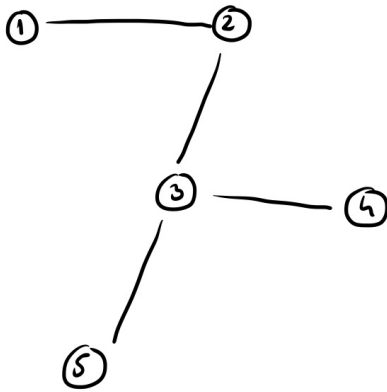
1 1



$$K \cdot N^3$$

$$\log K \cdot N^3$$

Matrix Multiplication



	1	2	3	4	5
1	0	1	0	0	0
2	1	0	1	0	0
3	0	1	0	1	1
4	0	0	1	0	0
5	0	0	1	0	0

= D

$$D^2 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

$$D^2 = \begin{pmatrix} \times & 0 & 1 & 0 & 0 \\ 0 & \times & 0 & 1 & \\ & & \times & & \\ & & & \times & \\ & & & & \times \end{pmatrix}$$

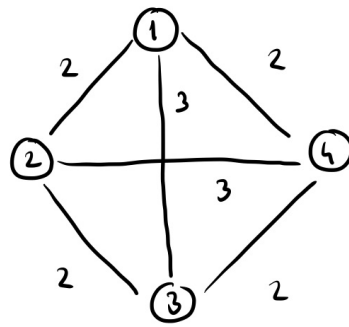
nu cont diag

Ignorăm elementele
de pe diag prin

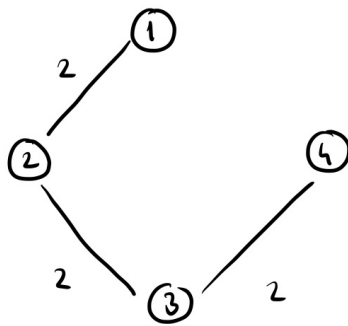
unde am 1 în D^2 înseamnă ca există drum de
lungime 2 de la i la j

$D^K =$ unde am 1 în D^K înseamnă ca există drum de
lungime K de la i la j

ex



APM:



$\neq$

Dijkstra

